

Process Vendor Invoice

Process Design Document (PDD)

Raya Information Technology
Authored by: Yasser Salah Mohamed

The logo for Raya Information Technology, featuring the letters 'RAYA' in a bold, blue, sans-serif font. The 'A's are stylized with a small triangle at the top. The logo is set against a white square background, which is placed on a grey rectangular area at the bottom right of the page.

Process Design Document History

Date	Version	Role	Name	Organization	Function	Comment
31.08.2025	1.0	Author	Mohamed Mesbah	ACME Systems Inc.	SME	Creation v 1.0
01.09.2025	1.1	Reviewer	Yasser Salah	Raya	BA	Approved v 1.0

Table of Contents

1.	Objective	4
1.1	Business Objectives and Benefits.....	4
2.	Current Process	4
2.1	Process Overview	4
2.2.	Standard of Metrics	5
2.3.	Volume of data.....	5
3.	Business Rules	6
4.	High Level Process Diagram	6
5.	Detailed Process Steps – AS IS	6
6.	In Scope for RPA.....	9
7.	Out of Scope.....	9
8.	Non-process requirements	10
8.1	Systems	10
8.2	Reporting.....	10
8.3	Archiving	10
8.4	Security & data retention.....	10
9.	Exception Handling	11
10.	Escalations.....	11

1. Objective

The objective of WI3 – Process Vendor Invoice in ACME System 1 is to ensure vendor invoices are received, validated, and recorded accurately in alignment with purchase orders and goods receipts, enabling timely payments and compliance with financial policies.

Automation within this process aims to:

- Reduce repetitive manual data entry.
- Minimize processing errors and mismatches.
- Accelerate invoice turnaround time.
- Improve transparency and compliance tracking.
- Free employees to focus on exception handling and higher-value activities.

1.1 Business Objectives and Benefits

Objectives	Expected Benefits
Eliminate manual data entry and comparison	Saves employee time and effort by reducing repetitive work.
Reduce human error in reconciliation	Improves accuracy and reliability of financial records.
Increase processing speed and consistency	Allows large volumes of accounts to be processed faster and in a standardized way.
Free employees for exception handling and value-added tasks	Enhances productivity and employee engagement by shifting focus to higher-value activities.
Ensure auditability of processed transactions	Provides consistent logs and evidence for compliance and audit purposes.

2. Current Process

2.1 Process Overview

The WI3 – Process Vendor Invoice in ACME System 1 covers the end-to-end handling of vendor invoices from receipt to validation and posting. The process ensures that invoice details are accurate, compliant, and aligned with purchase orders and goods receipts before being approved for payment. Automation supports the process by handling repetitive data entry, validation checks, and system updates, while employees manage exceptions and approvals.

2.2. Standard of Metrics

General information about the process selected for RPA, prior to automation:

Process name	Process Vendor Invoice
Function	Invoice Processing
Department	Finance / Accounting
Process short description	Handles vendor invoices in ACME System 1 by capturing details, validating against purchase orders and receipts, applying business rules, and posting for payment.
Process schedule	Runs daily on business days. Invoices are picked up from the ACME System 1 WI3 queue at scheduled intervals (e.g., every hour) until the working day ends.
#of items process/month	~ 2,000 – 2,500 invoices per month (average volume, may vary by vendor activity and business cycle).
Average cycle time	~ 5–7 minutes per invoice with automation.
Average handling time	~ 3–5 minutes per invoice for automated cases
Peak period(s)	End of each month, quarter, and fiscal year when vendor invoice submissions and payment processing volumes are highest.
# of FTE supporting this activity	~ 3 Full-Time Employee (FTEs) in the Accounts Payable team handle invoice processing, exception management, and approvals.
Input Data	Vendor Invoice Details (Primary document containing invoice details.)
Output Data	Validated and posted vendor invoice record in ACME System 1, updated work item status
Return on investment	Automating WI3 – Process Vendor Invoice reduces manual effort and errors, shortens processing time by over 50%, decreases FTE dependency, and delivers cost savings along with faster vendor payments and improved compliance.

2.3. Volume of data

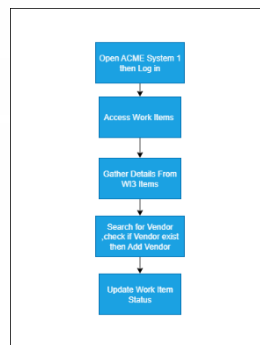
Activity	Time taken per month (hours)	Volume of data to be processed
Logging into ACME System 1	~5 hours	N/A (single login per session)
Invoice Retrieval (from WI3 queue)	~20 hours	~2,000–2,500 invoices
Data Extraction & Entry	~80 hours	~2,000–2,500 invoices
Validation (PO & GR match, tax, duplicates)	~100 hours	~2,000–2,500 invoices
Exception Handling (mismatches, missing data)	~40 hours	~300–400 invoices
Posting & Status Update in ACME System 1	~30 hrs	~2,000–2,500 invoices

Refers to the number of vendor invoices processed in ACME System 1, averaging 2,000–2,500 invoices per month, with ~300–400 falling into exception handling.

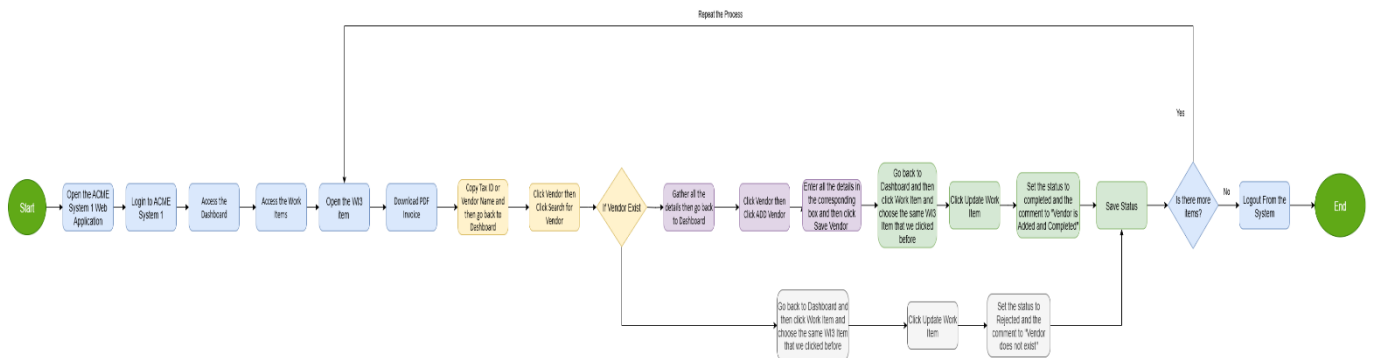
3. Business Rules

BR#	Requirement	Additional Information
#BR001	Duplicate invoices must be rejected.	Based on vendor ID + invoice number + amount check.
#BR002	Tax codes must be valid and correctly applied.	Validated against master data in ACME System 1.
#BR003	Vendor must exist in master data.	Invoices from unregistered vendors are flagged for review.
#BR005	All WI1 items must be processed within the same reporting cycle.	Typically daily, with higher volumes during month-end or quarter-end.
#BR006	User must log in with valid credentials before accessing work items.	Credentials are managed by the organization's IT/security policies.

4. High Level Process Diagram

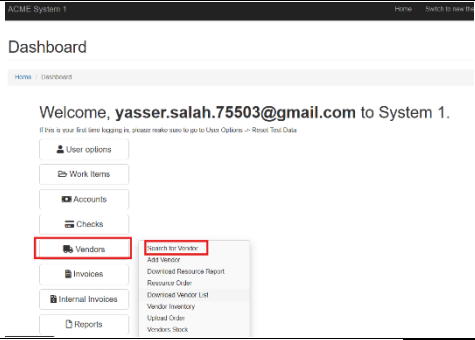
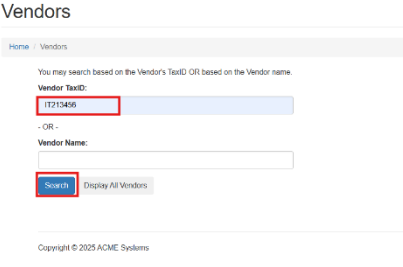
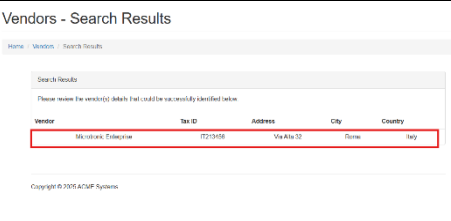
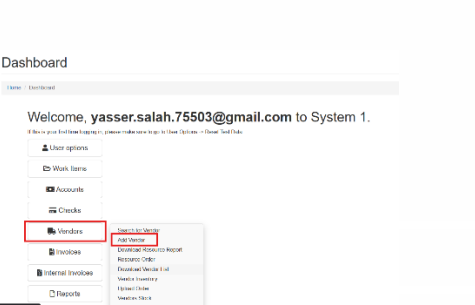
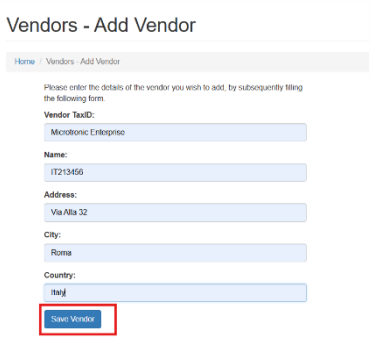
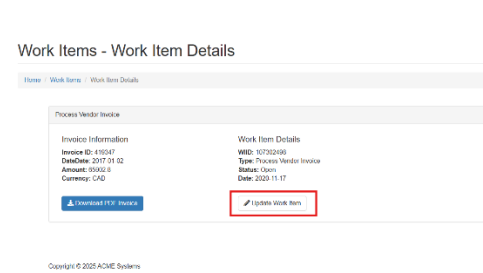


5. Detailed Process Steps – AS IS



Steps	Description	Screenshots	Exception Handling
-------	-------------	-------------	--------------------

1	Open the ACME System 1 Web Application		#SE001																																																																		
2	Log in to ACME System 1 using valid credentials.	Login Home / Login To continue, please authenticate here Email: Password: ☐ Remember Me Login Forgot Your Password? Register Copyright © 2025 ACME Systems	#SE003																																																																		
3	Access the Dashboard, it's the central location where the user can pick a specific menu item	ACME System 1 Home / Dashboard Dashboard Welcome, yasser.salah.75503@gmail.com to System 1. If this is your first time logging in, please make sure to go to User Options → Reset Test Data User options Work Items Accounts Checks Vendors Invoices Internal Invoices																																																																			
4	Access the Work Items Listing to consult all the available tasks to perform (Output data: task)	Work Items Home / Work Items Search Results Please find below your work items. They need to be completed in the order specified by your manager. <table><tr><th>Actions</th><th>WID</th><th>Description</th><th>Type</th><th>Status</th><th>Date</th></tr><tr><td> </td><td>10730001</td><td>Research Client Check Copy</td><td>W2</td><td>Open</td><td>2017-09-06</td></tr><tr><td> </td><td>10730002</td><td>Research Client Check Copy</td><td>W2</td><td>Open</td><td>2023-05-09</td></tr><tr><td> </td><td>10730003</td><td>Verify Account Problem</td><td>W1</td><td>Open</td><td>2019-03-26</td></tr><tr><td> </td><td>10730004</td><td>Calculate Client Security Check</td><td>W3</td><td>Open</td><td>2023-04-17</td></tr><tr><td> </td><td>10730005</td><td>Verify Account Problem</td><td>W1</td><td>Open</td><td>2017-09-13</td></tr><tr><td> </td><td>10730007</td><td>Research Client Check Copy</td><td>W2</td><td>Open</td><td>2021-09-16</td></tr><tr><td> </td><td>10730009</td><td>Calculate Client Security Check for Vendor</td><td>W3</td><td>Open</td><td>2023-04-23</td></tr><tr><td> </td><td>10730010</td><td>Verify Account Problem</td><td>W1</td><td>Open</td><td>2023-04-22</td></tr><tr><td> </td><td>10730014</td><td>Research Client Check Copy</td><td>W2</td><td>Open</td><td>2018-06-01</td></tr><tr><td> </td><td>10730033</td><td>Research Client Check Copy</td><td>W2</td><td>Open</td><td>2021-06-16</td></tr></table>	Actions	WID	Description	Type	Status	Date		10730001	Research Client Check Copy	W2	Open	2017-09-06		10730002	Research Client Check Copy	W2	Open	2023-05-09		10730003	Verify Account Problem	W1	Open	2019-03-26		10730004	Calculate Client Security Check	W3	Open	2023-04-17		10730005	Verify Account Problem	W1	Open	2017-09-13		10730007	Research Client Check Copy	W2	Open	2021-09-16		10730009	Calculate Client Security Check for Vendor	W3	Open	2023-04-23		10730010	Verify Account Problem	W1	Open	2023-04-22		10730014	Research Client Check Copy	W2	Open	2018-06-01		10730033	Research Client Check Copy	W2	Open	2021-06-16	#BE001
Actions	WID	Description	Type	Status	Date																																																																
	10730001	Research Client Check Copy	W2	Open	2017-09-06																																																																
	10730002	Research Client Check Copy	W2	Open	2023-05-09																																																																
	10730003	Verify Account Problem	W1	Open	2019-03-26																																																																
	10730004	Calculate Client Security Check	W3	Open	2023-04-17																																																																
	10730005	Verify Account Problem	W1	Open	2017-09-13																																																																
	10730007	Research Client Check Copy	W2	Open	2021-09-16																																																																
	10730009	Calculate Client Security Check for Vendor	W3	Open	2023-04-23																																																																
	10730010	Verify Account Problem	W1	Open	2023-04-22																																																																
	10730014	Research Client Check Copy	W2	Open	2018-06-01																																																																
	10730033	Research Client Check Copy	W2	Open	2021-06-16																																																																
5	Open WI3 item.	Search Results Please find below your work items. They need to be completed in the order specified by your manager. <table><tr><th>Actions</th><th>WID</th><th>Description</th><th>Type</th><th>Status</th><th>Date</th></tr><tr><td> </td><td>10730242</td><td>Research Client Check Copy</td><td>W2</td><td>Open</td><td>2023-05-09</td></tr><tr><td> </td><td>10730273</td><td>Calculate Client Security Check</td><td>W3</td><td>Open</td><td>2019-05-24</td></tr><tr><td> </td><td>10730292</td><td>Process Vendor Invoice</td><td>W3</td><td>Open</td><td>2023-12-01</td></tr><tr><td> </td><td>10730298</td><td>Process Vendor Invoice</td><td>W3</td><td>Open</td><td>2023-11-17</td></tr><tr><td> </td><td>10730344</td><td>Research Client Check Copy</td><td>W2</td><td>Open</td><td>2023-04-09</td></tr><tr><td> </td><td>10730360</td><td>Verify Account Problem</td><td>W1</td><td>Open</td><td>2023-04-20</td></tr><tr><td> </td><td>10730363</td><td>Verify Account Problem</td><td>W1</td><td>Open</td><td>2019-06-27</td></tr><tr><td> </td><td>10730364</td><td>Research Client Check Copy</td><td>W2</td><td>Open</td><td>2023-05-10</td></tr><tr><td> </td><td>10730398</td><td>Process Vendor Invoice</td><td>W3</td><td>Open</td><td>2023-12-21</td></tr><tr><td> </td><td>10730368</td><td>Verify Account Problem</td><td>W1</td><td>Open</td><td>2024-05-15</td></tr></table>	Actions	WID	Description	Type	Status	Date		10730242	Research Client Check Copy	W2	Open	2023-05-09		10730273	Calculate Client Security Check	W3	Open	2019-05-24		10730292	Process Vendor Invoice	W3	Open	2023-12-01		10730298	Process Vendor Invoice	W3	Open	2023-11-17		10730344	Research Client Check Copy	W2	Open	2023-04-09		10730360	Verify Account Problem	W1	Open	2023-04-20		10730363	Verify Account Problem	W1	Open	2019-06-27		10730364	Research Client Check Copy	W2	Open	2023-05-10		10730398	Process Vendor Invoice	W3	Open	2023-12-21		10730368	Verify Account Problem	W1	Open	2024-05-15	
Actions	WID	Description	Type	Status	Date																																																																
	10730242	Research Client Check Copy	W2	Open	2023-05-09																																																																
	10730273	Calculate Client Security Check	W3	Open	2019-05-24																																																																
	10730292	Process Vendor Invoice	W3	Open	2023-12-01																																																																
	10730298	Process Vendor Invoice	W3	Open	2023-11-17																																																																
	10730344	Research Client Check Copy	W2	Open	2023-04-09																																																																
	10730360	Verify Account Problem	W1	Open	2023-04-20																																																																
	10730363	Verify Account Problem	W1	Open	2019-06-27																																																																
	10730364	Research Client Check Copy	W2	Open	2023-05-10																																																																
	10730398	Process Vendor Invoice	W3	Open	2023-12-21																																																																
	10730368	Verify Account Problem	W1	Open	2024-05-15																																																																
6	For each activity of the type WI3 perform the following steps																																																																				
6.1	Click Download PDF Invoice	Work Items - Work Item Details Home / Work Items / Work Item Details Process Vendor Invoice Invoice Information Work Item Details Invoice ID: 419347 Date: 2017-01-02 Amount: 54919.0 Currency: USD WID: 10730292 Type: Process Vendor Invoice Status: Open Date: 2023-12-01 Download PDF Invoice Add New Work Item Copyright © 2025 ACME Systems																																																																			
6.2	Copy Tax ID, or Vendor Name to check if the vendor exist	INVOICE Vendor: Mecatronic Group Tax ID: RO094782 City: Buza Country: Romania Invoice ID: 419347 Date: 2017-01-02 TO: ACME Systems Inc. RO012326 Semenchere Road, 59 Building A Bucarest Romania <table><tr><th>Item #</th><th>Item Description</th><th>Unit Price</th><th>Units</th><th>Total</th></tr><tr><td>1</td><td>Various paper supplies</td><td>\$4919 CAD</td><td>1</td><td>\$4919 CAD</td></tr></table> Subtotal: 54919 CAD Tax: 10983.8 CAD	Item #	Item Description	Unit Price	Units	Total	1	Various paper supplies	\$4919 CAD	1	\$4919 CAD	#SE004																																																								
Item #	Item Description	Unit Price	Units	Total																																																																	
1	Various paper supplies	\$4919 CAD	1	\$4919 CAD																																																																	

6.3	Go back to the Dashboard and select Vendors then select search for vendor		
6.4	Enter the Tax ID, or the Vendor Name we Copied before and then click search		#BE004
6.5_Yes	If Vendor exist copy this details		#BE002
6.6_Yes	Go back to Dashboard ,then select Vendors and then select Add Vendor		
6.7_Yes	Enter the data we copied before and then click Save Vendor		
6.8_Yes	Go back to Dahboard then click Work Item then click the same WI3 Item we clicked before then click Update Work Item		

6.9_Yes	Set the status to Completed and the Comment to “Vendor is Added and Completed”. Then click Update Work Item	Update Work Item Update Work Item WIID: 107302498 Type: Process Vendor Invoice Status: Open Date: 2020-11-17 Comments: Add Comments: Vendor is Added and Completed New Status: Completed Update Work Item Copyright © 2025 ACME Systems						
6.5_No	If Vendor does not exist	Vendors - Search Results Home / Vendors / Search Results Search Results No vendor based on the name you have entered could be found in the database. Please contact the VDU team if you need to add a Vendor to the System <table><thead><tr><th>Vendor</th><th>Tax ID</th><th>Address</th><th>City</th><th>Country</th></tr></thead><tbody></tbody></table> Copyright © 2025 ACME Systems	Vendor	Tax ID	Address	City	Country	#BE003
Vendor	Tax ID	Address	City	Country				
6.6_No	Go back to Dahboard then click Work Item then click the same WI3 Item we clicked before then click Update Work Item	Work Items - Work Item Details Home / Work Items / Work Item Details Process Vendor Invoice Invoice Information Invoice ID: 419047 Date: 2017-01-02 Amount: 99002.8 Currency: CAD Download PDF Invoice Work Item Details WIID: 107302498 Type: Process Vendor Invoice Status: Open Date: 2020-11-17 Update Work Item Copyright © 2025 ACME Systems						
6.7_No	Set the status to Rejected and the Comment to “Vendor does not exist”. Then click Update Work Item	Update Work Item Update Work Item WIID: 107302498 Type: Process Vendor Invoice Status: Open Date: 2020-11-17 Comments: Add Comments: Vendor does not exist New Status: Rejected Update Work Item Copyright © 2025 ACME Systems						
7	If there more Items Continue with the next WI1 Activity							
8	Logout							

6. In Scope for RPA

The activities and exception in of scope for RPA, in this automation workflow /sequence are listed here:

- Full Scope of the process for RPA - to be 100% automated

7. Out of Scope

- There are no activities out of scope for RPA ...

8. Non-process requirements

8.1 Systems

System	Usage	Frequency
ACME System 1	Core system used for invoice retrieval (WI3), validation, posting, and reporting.	Daily (multiple times per day)

8.2 Reporting

Report name	Source	Purpose
Invoice Processing Report	ACME System 1	Provides summary of invoices processed (successful, failed, pending).
Exception Report	ACME System 1	Lists invoices with mismatches, missing data, or failed validations for manual review.
Compliance & Audit Report	ACME System 1	Ensures invoices are processed according to financial policies and audit standards.
Vendor Master Data Report	ACME System 1	Confirms vendor information is up-to-date for accurate invoice processing.

8.3 Archiving

All processed invoices and related reports are securely stored in ACME System 1 for future reference, compliance, and audit purposes, according to the company's data retention policy.

8.4 Security & data retention

Security & Data Retention: Access to invoice data is restricted to authorized users only, with role-based controls in ACME System 1. All processed invoices and related records are retained securely for a minimum of 7 years to comply with financial and audit regulations.

9. Exception Handling

9.1 Business Exception

#Business Exception no.	Business Exception	Description	Expected Action
#BE001	There is no work items	W13 Items does not exist	Raise exception for review; request to update the list
#BE002	Duplicate Invoice	Same vendor + invoice number already processed.	Reject or flag invoice; notify AP team.
#BE003	Invalid Vendor	Vendor not found in master data.	Route to master data team or reject invoice.
#BE004	Incorrect Tax Code	Tax details missing or invalid.	Escalate for finance review.
#BE005	Missing/Unreadable Data	Scanned invoice is unclear or incomplete.	Request resubmission from vendor.

9.2 System/Application Exception

# System Exception no.	Exception	Description	Expected Action
#SE001	System Unavailability	ACME System 1 is down or under maintenance.	Retry after system recovery; log downtime.
#SE002	Slow System Performance	Invoice queue takes longer to load or process.	Escalate to IT support; monitor queue backlog.
#SE003	Login/Authentication Failure	User or bot unable to log in due to invalid credentials or expired password.	Reset credentials; escalate to IT/security team.
#SE004	Data Extraction Error	Invoice data fields not captured correctly (OCR/formatting issue).	Re-run extraction; escalate if repeated.
#SE005	Timeout/Error Message	Session times out during validation or posting.	Restart process; log error; escalate if frequent.

10. Escalations

For immediate contacts for the process automation project, kindly contact the respective contact on the table below:

Email Address	Roles	Responsibilities
sme.ap@acme.com	Subject Matter Expert (SME)	Provides process knowledge, clarifies business rules, and assists in resolving complex invoice exceptions.
senior.dev@acme.com	Senior Developer	Handles automation-related issues, fixes workflow errors, and ensures bot stability and performance.
ba@acme.com	Business Analyst	Analyzes exception trends, validates requirements, and coordinates between business users, SMEs, and developers for resolution.